

Boundary-Trace Fixation under Admissibility

Forced Environmental Coupling as the Final Role-Anchor Asymmetry

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Abstract

This paper supplies the Boundary-Trace Fixation theorem for AASC: the missing fixed-domain result governing role-anchor fixation under constructional non-selectivity. The target problem is not confirmation but fixation: a construction system contains multiple candidate tokens for the same constructional role, while internal structure, admissible outcome-orderings, provenance, and indexical anchoring fail to select a unique occupant. The claim is not that environmental facts directly determine reference. The claim is narrower and stronger: once internal and indexical/provenance asymmetries are exhausted, a unique declared boundary trace is the only remaining admissible asymmetry capable of fixing the role.

The environmental move is typed as a boundary-trace construction. The selector does not consult the environment as an oracle. It consults only a bivalent, declared, channel-preserving predicate represented inside the target construction domain. This blocks illicit external injection: externality is admissibility-relevant only when fixed as a boundary-status invariant of the evaluated domain. An asymmetry-source normal form is proved: every putative role-fixing selector is either a same-domain invariant, a declared indexical/provenance marker, a declared boundary-coupling trace, or an inadmissible extra-domain preference. Exhaustion then follows from the normal form rather than from taxonomy.

The main theorem states that if a construction system is outcome-complete for the role-assignment problem, internally non-selective, and indexically unanchored, then a singleton stable boundary trace fixes the constructional role-anchor. Absence, multiplicity, instability, outcome-conditioning, scalarization, delayed calibration, and post-selection repair fail closed. A compact structural witness and executable witness are supplied. The result is a fixation theorem for admissible environmental construction-anchoring, not an empirical confirmation rule or an interpretive convention.

One-line gloss. Under admissibility and internal non-selectivity, a unique role-anchor exists iff a singleton stable boundary trace exists; otherwise fixation fails closed.

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1 Scope, terminology update, and proof posture

Reader orientation. A minimal working understanding is enough to follow the argument: AASC treats an admissible construction as a fixed-domain identity-preserving construction whose standing is bivalent and cannot be generated, transferred, or repaired after failure. In that setting, the present paper asks what can still fix a constructional role when internal constructional resources do not select a unique token.

Definition 1.1 (Minimal terminology update). The terminology used here is normalized as follows. The legacy phrase “evidential role” is treated as *constructional role*; “record token” is treated as *constructed token with record function*; “reference assignment” is treated as *reference-fixing role-anchor assignment*; “prediction” is treated as *outcome profile* unless an ordinary empirical gloss is intended; and “minimal witness” is treated as a *construction witness*. The term “reference” is retained only for the fixed task of assigning standing to the token that occupies the constructional role.

The target problem is not how to confirm a constructed alternative. It is earlier and logically prior: how a construction system fixes which internal token occupies the constructional role whose reports will later be used in confirmation. This is the fixation layer of the AASC architecture. The construction problem isolates one route: a candidate token may be stably coupled to an external

environment in a way that the other candidates are not. This paper makes that route precise enough to withstand the standard AASC objections.

The result is upgraded here from a structural existence theorem to a forced fixed-domain fixation theorem. It states that a unique, stable, outcome-independent boundary trace induced by an environmental channel can be an admissible construction-anchor. It does not state that every causal relation, every sensor relation, or every environmental variable fixes reference. It also does not make environmental coupling a new confirmation score. The coupling is admissible only when it is fixed prior to the confirmational use of the record and when it is preserved under the transformations that count as identity-preserving on the declared environmental domain.

Definition 1.2 (Target fixed domain). The target fixed domain is

$$\mathcal{D}_{\text{env}} = (\mathcal{C}, \mathcal{L}, \mathcal{S}, \text{Rep}, \text{Out}, \text{Adm}, \text{Aut}, \mathcal{E}, \text{Ch}),$$

where:

- (i) $\mathcal{C}$ is the construction system under evaluation;
- (ii) $\mathcal{L}$ is a finite scheme of constructional roles;
- (iii) $\mathcal{S}$ is the class of constructed internal token candidates and other internal structures of $\mathcal{C}$;
- (iv) **Rep** and **Out** are the report-content and outcome-profile interfaces induced by a role assignment;
- (v) **Adm** is the fixed-domain admissibility predicate for role assignments;
- (vi) **Aut** is the class of identity-preserving transformations preserving the fixed-domain construction structure under evaluation;
- (vii) $\mathcal{E}$ is an external environment declared as part of the target environmental domain; and
- (viii) **Ch** is the declared family of interaction channels by which constructed tokens in $\mathcal{S}$ may or may not be coupled to $\mathcal{E}$.

The domain is *fixed* only after these ingredients are declared. Adding a new environment, changing the channel family, changing the role vocabulary, or changing the admissibility predicate is a scope change, not a same-scope repair.

Remark 1.3 (Why the environment is not smuggled in). An environmental channel used in Definition 1.2 is not introduced after a confirmational failure in order to rescue a preferred result. It is part of the declared environmental domain. If the channel was not present in the domain on which the original reference question was posed, then the correct AASC description is not “same-domain repair”; it is explicit domain reset. The theorem below is conditional on the reset domain $\mathcal{D}_{\text{env}}$ and makes no claim that an undeclared environment can retroactively repair reference failure inside $\mathcal{D}_{\text{int}}$.

Definition 1.4 (AASC import discipline). An import from the AASC corpus is used here only as a fixed-domain structural discipline:

- (1) reference-stable invariants must be preserved under identity-preserving transformation;
- (2) admissibility is bivalent on the evaluated fragment;
- (3) the pre-admissible boundary admits no meaningful selector, score, metric, or ranking among otherwise symmetric candidates;

- (4) standing and admissibility cannot be repaired after failure as the same evaluated act;
- (5) scope changes must be declared rather than silently used as same-scope evidence; and
- (6) admissible role-anchor fixation is required before a role-indexed constructed token can carry theory-level confirmational work.

Appendix A records the AASC source labels used by the proof spine.

Remark 1.5 (Positive result and no-go result). The paper proves both sides. The positive theorem identifies a sufficient environmental invariant. The converse and exhaustion results show that the positive theorem is sharp within the declared same scope: once the environmental invariant is not unique, not stable, not outcome-independent, or not in scope, the missing reference cannot be supplied by tie-breaking, posterior success, convention, scalar coupling strength, delayed calibration, or relabeling.

2 Internal constructional role-anchor problem

Let $\mathcal{H} = \{H_0, H_1\}$ be the set of theory-internal constructed alternatives under evaluation. The binary form is not essential; it is retained because it makes the minimal construction witness transparent. Let $\ell \in \mathcal{L}$ be the constructional role to be fixed, for example “the constructed token whose content is used as evidence.” Let

$$\mathcal{S}_\ell \subseteq \mathcal{S}, \quad |\mathcal{S}_\ell| \geq 2,$$

be the candidate set of constructed internal tokens that can occupy ℓ absent further role-anchoring structure.

Definition 2.1 (Role assignment). A role assignment is a partial map

$$R : \mathcal{L} \rightarrow \mathcal{S}.$$

For $s \in \mathcal{S}_\ell$, write $R[\ell \mapsto s]$ for the assignment obtained by assigning the constructional role ℓ to token s while leaving all other assigned roles fixed.

Definition 2.2 (Report and outcome-profile interfaces). For a role assignment R , let $\text{Rep}(R)$ be the total report content the construction system treats as available under that assignment, and let $\text{Out}(R)$ be the outcome-profile family assigned to the admissible contexts of use under that assignment. Equality of Rep means equality of report content, not merely equality of inscriptions. Equality of Out means equality of the admissibility-relevant outcome profile over all contexts counted as meaningful on the fixed domain. The notation Out is retained only as legacy mathematical shorthand for the outcome-profile interface.

Definition 2.3 (Outcome-complete construction system). A construction system $\mathcal{C}$ is *outcome-complete relative to the role-assignment problem* $(\ell, \mathcal{S}_\ell)$ iff every candidate assignment $R[\ell \mapsto s]$ with $s \in \mathcal{S}_\ell$ has a fixed report-content interface and a fixed outcome-profile interface on the declared domain, relative to all admissible outcome-orderings over $\mathcal{H}$ available in that domain. Outcome-completeness does not mean that the system is empirically true or predictively successful. It means that every admissible outcome-ordering already assigns the same role-relevant outcome profile to the candidates, so outcome comparison cannot itself supply an additional role selector.

Definition 2.4 (Outcome-profile neutrality). The candidate set $\mathcal{S}_\ell$ is outcome-profile neutral for ℓ iff for all $s, t \in \mathcal{S}_\ell$ and for every admissible outcome-ordering in the declared domain,

$$\text{Out}(R[\ell \mapsto s]) = \text{Out}(R[\ell \mapsto t])$$

whenever both assignments are physically consistent on the fixed internal domain. Older wording would call this predictive indistinguishability; the operative AASC content is neutrality of the role-indexed outcome profile across admissible outcome-orderings.

Definition 2.5 (Internal non-selectivity). The construction system $\mathcal{C}$ is internally non-selective with respect to $(\ell, \mathcal{S}_\ell)$ iff no feature definable purely within the internal formal structure of $\mathcal{F}$ privileges any $s \in \mathcal{S}_\ell$ as the unique occupant of ℓ while preserving the fixed report and outcome-profile interfaces. Equivalently, any proposed internal selector either:

- (i) fails to be unique;
- (ii) changes report content or outcome-profile content;
- (iii) depends on labels, coordinates, or representatives rather than fixed-domain structure; or
- (iv) imports additional structure not contained in the internal domain.

Definition 2.6 (Admissible structure automorphism). An admissible structure automorphism of the internal candidate problem is a bijection

$$\sigma : \mathcal{S} \rightarrow \mathcal{S}$$

such that, for every admissible role assignment R :

- (i) $\sigma \circ R$ is admissible whenever R is admissible;
- (ii) $\text{Rep}(\sigma \circ R) = \text{Rep}(R)$;
- (iii) $\text{Out}(\sigma \circ R) = \text{Out}(R)$; and
- (iv) σ preserves the fixed identity conditions for the internal structures under evaluation.

The class of such transformations is denoted $\text{Aut}_{\text{int}}(\mathcal{F})$.

Definition 2.7 (Symmetry saturation). The candidate set $\mathcal{S}_\ell$ is internally symmetry-saturated iff for every $s, t \in \mathcal{S}_\ell$ there exists $\sigma \in \text{Aut}_{\text{int}}(\mathcal{F})$ with $\sigma(s) = t$. It is environmentally symmetry-saturated relative to a coupling predicate $C : \mathcal{S}_\ell \rightarrow \{0, 1\}$ iff the channel-preserving automorphisms leave no singleton positive orbit: for every $s \in \mathcal{S}_\ell$ with $C(s) = 1$, either there is another $t \neq s$ with $C(t) = 1$, or no candidate has $C = 1$.

Lemma 2.8 (Internal neutrality blocks internal selection). *If $\mathcal{F}$ is internally non-selective with respect to $(\ell, \mathcal{S}_\ell)$ and $\mathcal{S}_\ell$ is internally symmetry-saturated, then no same-scope internal rule uniquely selects a candidate for ℓ without violating admissibility.*

Proof. A unique internal rule would select some $s^* \in \mathcal{S}_\ell$. Since $\mathcal{S}_\ell$ is internally symmetry-saturated, there is an admissible internal automorphism carrying s^* to another candidate t . The rule can remain same-scope and admissible only if it is invariant under this transformation. Invariance forces it to select both s^* and t , contradicting uniqueness. If instead the rule selects s^* but not t , then it is not preserved by an admissible symmetry and therefore depends on a label, representative, or additional structure not in the internal domain. That is exactly the forbidden selector route. $\square$

Remark 2.9 (Why outcome comparison is not enough). Outcome-profile neutrality is not a claim that empirical outcomes are unimportant. It says that outcome comparison does not perform the role-anchoring work in the case under study. If two candidate role assignments preserve all outcome profiles and report content available internally, then later success cannot distinguish which candidate was the constructional-role occupant. A confirmational rule may operate after reference is fixed; it cannot be the same-scope source of the fixation without becoming post-selection repair or outcome-conditioned selection.

3 Kernel-to-interface-to-forcing chain

The result becomes closure-class only after the dependency chain is made explicit. The relevant AASC chain is:

Adm $\Rightarrow$ fixed-domain identity preservation
 $\Rightarrow$ reference-stable invariants
 $\Rightarrow$ internal non-selectivity diagnosis
 $\Rightarrow$ asymmetry exhaustion
 $\Rightarrow$ forced environmental anchoring when the other routes fail.

The point is not that environmental coupling is always available. The point is that, once the reference question is posed on a fixed domain and the internal and indexical routes are unavailable, any remaining admissible same-scope selector must be a stable declared boundary invariant. Class III environmental coupling is exactly that type: the only remaining admissible asymmetry once internal and indexical asymmetries are exhausted.

Definition 3.1 (Role-anchor asymmetry type). A role-anchor asymmetry type for $(\ell, \mathcal{S}_\ell)$ is a same-domain invariant family A such that, for some candidate $s^* \in \mathcal{S}_\ell$, the statement “ s^* is the unique occupant of ℓ ” is fixed by A and preserved under every identity-preserving transformation in the declared domain. An asymmetry type is admissible only if it is declared before confirmational use, bivalent at the point of role fixation, and not introduced by relabeling, posterior success, tuning, or post hoc repair.

Definition 3.2 (The three admissible asymmetry loci). For a fixed role-assignment problem, admissible asymmetry can occur only at one of three loci:

- (i) **internal constructional structure** lnt: an invariant definable wholly inside the fixed internal construction tuple;
- (ii) **indexical/provenance anchoring** ldx: a declared origin, index, or production-history invariant already in the target domain; or
- (iii) **environmental boundary trace** Tr_{Ch} : a declared stable trace induced by a channel between a constructed token and the environmental boundary domain.

A proposed fourth locus is not a same-domain admissible asymmetry unless it reduces to one of these three or explicitly resets the domain.

Lemma 3.3 (Asymmetry source normal form). *Every putative same-domain selector for a constructional role falls into one of four and only four normal-form classes:*

- (N1) *Same-domain invariant.* The selector is fixed by structure already internal to $(\mathcal{C}, \mathcal{S}, \text{Rep}, \text{Out}, \text{Adm})$.
- (N2) *Declared indexical/provenance marker.* The selector is fixed by an origin, production-history, indexical, or centered marker declared in the target domain before role use.
- (N3) *Declared boundary-coupling trace.* The selector is fixed by a bivalent boundary-status trace induced by a declared channel between a constructed token and an external environment included in the target domain.
- (N4) *Inadmissible extra-domain preference.* The selector depends on pragmatic salience, policy, convenience, semantic intention, temporal priority, causal ancestry, metadata, ranking, scalar tuning, later success, or another preference not fixed as an admissibility-preserving invariant of the declared domain.

Cases (N1)–(N3) are the only admissible asymmetry-source forms. Case (N4) is not a fourth admissible source; if declared as new structure it is a scope reset, and if left undeclared it is an illicit same-scope selector.

Proof. A same-domain selector must fix a unique role occupant by some feature preserved under identity-preserving transformations. If the feature is internal to the construction system, it is (N1). If it is a declared marker of origin, position, or production history, it is (N2). If it is neither internal nor indexical/provenance-based but remains declared in the target domain, the only remaining admissible location is the boundary between a constructed token and declared extra-internal structure; because role fixation is bivalent, that boundary can enter only as a bivalent trace, giving (N3). All other candidates either reduce to one of these after explicit declaration or else decide the role by a preference not invariantly fixed in the domain. That is (N4). The four cases are mutually exclusive at the decisive selector locus and exhaustive over the locations from which a same-domain selector can draw authority. $\square$

Theorem 3.4 (Asymmetry exhaustion for role fixation). *Let $(\ell, \mathcal{S}_\ell)$ be a fixed-domain constructional-role problem with $|\mathcal{S}_\ell| \geq 2$. Suppose an admissible construction fixes a unique occupant of ℓ . Then the decisive asymmetry must be internal constructional, indexical/provenance-based, or boundary-trace environmental in the sense of Lemma 3.3 and definition 3.2. By the normal-form lemma, no admissible selector exists outside these three classes. If all admissible normal-form classes (N1)–(N3) are unavailable or non-singleton on $\mathcal{S}_\ell$, no same-domain admissible construction fixes ℓ .*

Proof. By Lemma 3.3, every putative same-domain selector is either a same-domain invariant, a declared indexical/provenance marker, a declared boundary-coupling trace, or an inadmissible extra-domain preference. The first three classes are exactly the loci listed in Definition 3.2. The fourth class is not an admissible locus: if the preference is declared as new fixed structure, the evaluated domain has changed; if it is not declared, the selector is silent redescription, post-selection repair, parameter injection, or extra-domain policy. Therefore an admissible role-fixing construction must draw its decisive asymmetry from one of the three admissible loci. Equivalently, by the normal-form lemma, no admissible selector exists outside these classes. If all three fail uniqueness or availability, no same-domain admissible construction fixes ℓ . $\square$

Table 1: Exhaustive normalization of admissible asymmetry sources. Each candidate either reduces to an admissible normal-form class under explicit declaration, or fails as an extra-domain preference, parameter injection, or post-selection repair.

Candidate source	asymmetry	Normal-form collapse	Admissibility result
Temporal priority		Same-domain invariant if the ordering is declared; otherwise extra-domain preference	Admissible only when the temporal ordering is part of the fixed domain
Causal ancestry		Boundary trace or provenance marker if declared; otherwise extra-domain causal story	Admissible only through declared channel/provenance typing
Semantic intention		Declared indexical/provenance marker, or else extra-domain intention	Admissible only when fixed before role use
Agent policy		Extra-domain preference unless encoded as fixed domain structure, which resets the domain	Not a same-scope selector
Model convenience		Extra-domain preference	Inadmissible
Pragmatic salience		Extra-domain preference	Inadmissible
Metadata		Internal invariant, provenance marker, or boundary trace if declared; otherwise extra-domain preference	Admissible only after normal-form typing
Scalar ranking		Boundary trace only after declared bivalent threshold; otherwise parameter injection	Inadmissible unless reduced to a fixed bivalent predicate
Later success		Post-selection repair or fresh-domain evaluation	No same-act repair

Theorem 3.5 (Internal selector impossibility under constructional non-selectivity). *If $\mathcal{C}$ is internally non-selective with respect to $(\ell, \mathcal{S}_\ell)$ and $\mathcal{S}_\ell$ is outcome-profile neutral for ℓ , then no internally definable same-domain construction operator uniquely fixes ℓ without violating fixed-domain admissibility.*

Proof. Assume an internal operator Y uniquely fixes ℓ by selecting $s^* \in \mathcal{S}_\ell$. Since Y is internal, it must be definable from the internal constructional structure of $\mathcal{C}$ together with the fixed report and outcome-profile interfaces. Internal non-selectivity states precisely that no such internal feature privileges a unique occupant while preserving those interfaces. Outcome-profile neutrality removes outcome comparison as a distinguishing invariant. Therefore Y must use something not present in the internal constructional structure, or it must relabel a symmetric feature as decisive. The first case leaves the internal domain; the second is silent redescription. Either way, Y is not an admissible internal same-domain operator. $\square$

Corollary 3.6 (Forcing of environmental coupling after internal and indexical exhaustion). *Let $(\ell, \mathcal{S}_\ell)$ be internally non-selective and outcome-profile neutral. If no internal constructional asymmetry and no indexical/provenance anchor is available on $\mathcal{S}_\ell$, then any admissible same-domain role-fixing construction must be a declared environmental-coupling invariant. If no singleton environmental-coupling invariant exists, the role remains unfixed.*

Proof. By Theorem 3.4, any admissible asymmetry must be internal, indexical/provenance-based, or environmental. By Theorem 3.5, the internal route is closed under the stated hypotheses. By assumption, the indexical/provenance route is closed. The only remaining same-domain asymmetry locus is environmental coupling. If that locus lacks a singleton stable invariant, it cannot fix a unique role occupant. $\square$

4 Environmental coupling as declared channel structure

Environmental coupling is a positive construction-anchor only when it is represented as a fixed boundary trace induced by a declared channel relation, not when it is a score assigned after the fact. The distinction between those two uses is the core of the theorem.

Definition 4.1 (External environment). An external environment $\mathcal{E}$ is a system not represented exhaustively by the internal constructional formalism of $\mathcal{C}$, but included in the declared target domain $\mathcal{D}_{\text{env}}$ as an interaction partner with which constructed internal constructed tokens may stand in channel-preserving relations.

Definition 4.2 (Interaction channel). An interaction channel for a token $s \in \mathcal{S}$ is a fixed relation

$$\text{Ch}(s, \mathcal{E})$$

whose presence or absence is part of the declared environmental domain. Examples include persistent sensor linkage, calibration lock, continuous causal interface, externally maintained synchronization, or a physically declared readout channel. The examples are illustrative; the formal role is channel preservation, not a particular technology.

Definition 4.3 (Typed declared boundary trace). A declared boundary trace for $(\ell, \mathcal{S}_\ell)$ is the fixed-domain boundary-status predicate

$$\text{Tr}_{\text{Ch}} : \mathcal{S}_\ell \longrightarrow \{0, 1\},$$

where

$$\text{Tr}_{\text{Ch}}(s) = 1 \iff s \text{ is stably coupled to } \mathcal{E} \text{ through the declared channel family } \text{Ch}.$$

Its valuation is invariant under every admissible transformation that preserves the declared channel package:

$$\sigma \in \text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}}) \implies \text{Tr}_{\text{Ch}}(\sigma(s)) = \text{Tr}_{\text{Ch}}(s) \quad \text{for all } s \in \mathcal{S}_\ell.$$

The trace is part of the evaluated construction state, not an external query. It is non-generative: it introduces no new candidate tokens and changes no report content or outcome-profile interface. It is standing-preserving exactly when its valuation is bivalent, declared before role use, stable under channel-preserving transformations, and independent of confirmational outcome-orderings. The construction system does not inspect $\mathcal{E}$ directly as an oracle and does not use an unrestricted environmental fact as a selector. It uses only the declared trace value fixed by the channel package in $\mathcal{D}_{\text{env}}$.

Lemma 4.4 (Boundary trace lemma). *Environmental coupling can enter an admissible role-anchor assignment only as a typed declared boundary trace. Direct appeal to the external environment itself is not a same-domain selector.*

Proof. The environment may contain facts not represented exhaustively by the internal constructional formalism. Those facts do not automatically become admissibility-relevant. A same-domain selector must be typed in the target domain and preserved by the transformations that define identity for that domain. The object satisfying those conditions is not $\mathcal{E}$ as an external ontology; it is the bivalent trace $\text{Tr}_{\text{Ch}} : \mathcal{S}_\ell \rightarrow \{0, 1\}$ declared by the channel package. If a selector consults facts of $\mathcal{E}$ outside Tr_{Ch} , then it either adds new structure to the domain or uses an undeclared external preference. The former is a scope reset; the latter is inadmissible. Because the trace is typed on $\mathcal{S}_\ell$, it cannot generate a token, change the role, or import a new outcome-ordering. Its only admissible function is to preserve or reject a candidate role-anchor assignment according to a declared boundary-status invariant. Therefore admissible environmental anchoring is boundary-trace anchoring. $\square$

Definition 4.5 (Channel-preserving transformation). An admissible transformation $\sigma \in \text{Aut}_{\text{int}}(\mathcal{F})$ is channel-preserving on $\mathcal{D}_{\text{env}}$ iff it preserves the declared environmental channel package:

$$\text{Tr}_{\text{Ch}}(\sigma(s)) = \text{Tr}_{\text{Ch}}(s) \quad \text{for all } s \in \mathcal{S}$$

whenever σ is claimed to be a same-domain transformation of $\mathcal{D}_{\text{env}}$. The channel-preserving subgroup is denoted

$$\text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}}) \subseteq \text{Aut}_{\text{int}}(\mathcal{F}).$$

Definition 4.6 (Stability of coupling). The boundary trace Tr_{Ch} is stable on $\mathcal{D}_{\text{env}}$ iff, across all admissible same-domain transformations, coupling status is preserved:

$$\forall \sigma \in \text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}}), \forall s \in \mathcal{S}, \quad \text{Tr}_{\text{Ch}}(\sigma(s)) = \text{Tr}_{\text{Ch}}(s).$$

Equivalently, coupledness is a reference-stable invariant of the environmental domain.

Definition 4.7 (Outcome independence). A role-anchor constraint X is outcome-independent iff its value does not depend on:

- (i) which constructed alternative in $\mathcal{H}$ is favored;

- (ii) posterior values, likelihood ratios, Bayes factors, losses, utilities, or decision scores;
- (iii) confirmational ordering over constructed alternatives; or
- (iv) downstream success, failure, or explanatory convenience.

For environmental coupling, this means that Tr_{Ch} is fixed by the channel package Ch and not by any later value of Conf , posterior update, or outcome-profile ranking.

Definition 4.8 (Admissible environmental anchor). A boundary trace Tr_{Ch} is an admissible environmental anchor for $(\ell, \mathcal{S}_\ell)$ iff it satisfies:

- (i) it is well-typed in the declared target domain $\mathcal{D}_{\text{env}}$;
- (ii) it is stable under $\text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}})$;
- (iii) it is outcome-independent;
- (iv) it is bivalent as a coupling-status predicate, $\text{Tr}_{\text{Ch}} : \mathcal{S}_\ell \rightarrow \{0, 1\}$; and
- (v) its use does not alter Rep or Out except by fixing which already available token occupies ℓ .

Remark 4.9 (Bivalent predicate, not scalar coupling strength). A real physical setup may contain degrees of coupling strength. The admissibility-relevant environmental anchor used here is the boundary trace, not a scalar score over such degrees. Either the declared channel criterion is satisfied or it is not. If a scalar threshold is introduced as the decisive boundary criterion, then the threshold must itself be part of the declared domain; otherwise it is parameter injection. Once declared, only the bivalent predicate induced by that threshold is used for role-anchor fixation.

The next lemma specializes Lemma 4.4 to the environmental case by listing the admissibility clauses used in the main theorem.

Lemma 4.10 (Admissibility of declared environmental coupling). *A boundary trace $\text{Tr}_{\text{Ch}} : \mathcal{S}_\ell \rightarrow \{0, 1\}$ is an admissible construction anchor only if it satisfies all of the following: it is declared in $\mathcal{D}_{\text{env}}$ before role use, it is bivalent on $\mathcal{S}_\ell$, it is stable under $\text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}})$, it is outcome-independent, and it does not alter Rep or Out when used to assign ℓ . Under these conditions, using Tr_{Ch} to restrict role assignment is not illicit external injection.*

Proof. The alleged illicit-injection worry arises only if the environment is used as an oracle, if the environment is added to a domain after reference failure, or if the channel changes the constructed object whose standing is being evaluated. Declaration in $\mathcal{D}_{\text{env}}$ blocks the first failure: the channel is part of the evaluated domain, not an after-the-fact rescue. Bivalence and stability make the channel a fixed invariant rather than a tunable score. Outcome independence blocks posterior or success-conditioned selection. Preservation of Rep and Out blocks constructional mutation of the token. Therefore the trace functions as a boundary condition on admissible role assignment, not as a new outcome, an environmental oracle, or a hidden selector. $\square$

Lemma 4.11 (Unique coupling fixes the coupled token under channel-preserving symmetry). *Suppose $\text{Tr}_{\text{Ch}} : \mathcal{S}_\ell \rightarrow \{0, 1\}$ is stable and there exists exactly one $s^* \in \mathcal{S}_\ell$ such that $\text{Tr}_{\text{Ch}}(s^*) = 1$. Then every channel-preserving admissible transformation fixes s^* :*

$$\forall \sigma \in \text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}}), \quad \sigma(s^*) = s^*.$$

Proof. Let $\sigma \in \text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}})$. By stability,

$$\text{Tr}_{\text{Ch}}(\sigma(s^*)) = \text{Tr}_{\text{Ch}}(s^*) = 1.$$

By uniqueness, s^* is the only candidate in $\mathcal{S}_\ell$ with boundary-trace value 1. Therefore $\sigma(s^*) = s^*$. $\square$

Lemma 4.12 (Environmental coupling is not evidence-based symmetry breaking). *If Tr_{Ch} is fixed by Ch and satisfies outcome independence, then environmental selection does not break internal symmetry on the basis of confirmational evidence.*

Proof. The value of $\text{Tr}_{\text{Ch}}(s)$ is determined by the declared channel relation between s and $\mathcal{E}$. By outcome independence, changing which constructed alternative is favored, which posterior is larger, or which outcome succeeds does not change Tr_{Ch} . Therefore environmental selection is not a function of confirmational evidence. It is a channel-invariant restriction on admissible role assignment. $\square$

5 Boundary-trace fixation rule

An argmax presentation is insufficient for the present theorem. The admissible formulation is a boundary-trace selector with an explicit fail-closed condition. Since the boundary trace Tr_{Ch} is bivalent, an argmax alone is not enough: if two candidates have value 1, the argmax set is non-singleton; if no candidate has value 1, the argmax over zeros would select everything unless the rule blocks that case. The admissible rule is therefore a partial function.

Definition 5.1 (Boundary-trace fixation operator). For a fixed role-assignment problem $(\ell, \mathcal{S}_\ell)$, the boundary-trace fixation operator is the partial operator

$$\text{BTFix}_{\text{Ch}} : (\ell, \mathcal{S}_\ell) \rightarrow \mathcal{S}_\ell$$

defined exactly by singleton positivity of the declared trace:

$$\text{BTFix}_{\text{Ch}}(\ell, \mathcal{S}_\ell) = s \quad \text{iff} \quad \{t \in \mathcal{S}_\ell : \text{Tr}_{\text{Ch}}(t) = 1\} = \{s\}.$$

If this singleton condition fails, the operator is undefined. Thus fixation is not a choice procedure over symmetric candidates; it is the preservation of the unique candidate already fixed by the declared boundary trace.

Definition 5.2 (Environmental selection rule). For a candidate set $\mathcal{S}_\ell$, define the environmental presentation of the boundary-trace fixation operator by

$$X_{\text{env}}(\mathcal{S}_\ell) = \text{BTFix}_{\text{Ch}}(\ell, \mathcal{S}_\ell) = \begin{cases} s, & \text{if } \{t \in \mathcal{S}_\ell : \text{Tr}_{\text{Ch}}(t) = 1\} = \{s\}, \\ \uparrow, & \text{otherwise.} \end{cases}$$

Here $\uparrow$ means fail-closed undefinedness, not a third admissibility status and not a prompt for tie-breaking.

Definition 5.3 (Environmental role-anchor assignment). If $X_{\text{env}}(\mathcal{S}_\ell) = s^*$, the environmental role-anchor assignment for constructional role ℓ is

$$R_{\text{env}} := R[\ell \mapsto s^*].$$

If $X_{\text{env}}(\mathcal{S}_\ell) = \uparrow$, no environmental role-anchor assignment is produced on the fixed scope.

Lemma 5.4 (Fail-closedness blocks selector smuggling). *The partiality of X_{env} is necessary. If X_{env} returned a candidate when the coupled set had cardinality 0 or at least 2, then the rule would introduce an additional selector not supplied by the environmental coupling predicate.*

Proof. If the coupled set is empty, the boundary trace supplies no positive candidate. Any returned token is therefore selected by some other criterion. If the coupled set has at least two elements, the boundary trace treats those elements alike. Selecting one of them requires an additional tie-breaker. In either case the returned value is not determined by Tr_{Ch} alone. Unless the extra criterion is declared as new domain structure, it is a hidden same-scope selector; if it is declared, the original domain has changed. $\square$

Proposition 5.5 (Environmental assignment is reference-stable). *If Tr_{Ch} is an admissible environmental anchor and $X_{\text{env}}(\mathcal{S}_\ell) = s^*$, then the assignment R_{env} is stable under channel-preserving admissible transformations with respect to the role ℓ .*

Proof. By Lemma 4.11, every transformation in $\text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}})$ fixes s^* . Therefore every such transformation preserves the statement that ℓ is assigned to s^* . Since the rule does not modify the remaining assignment structure except by fixing ℓ , R_{env} is stable at the reference-fixing locus. $\square$

Remark 5.6 (The rule is positive only in the singleton case). The positive Class III case is not “environmental information is available.” The positive case is narrower: the boundary trace must have a unique positive token among the candidates. Environmental richness without singleton asymmetry is not enough.

6 Main theorem: Boundary-Trace Fixation

Theorem 6.1 (Boundary-Trace Fixation Theorem). *Let $\mathcal{C}$ be outcome-complete relative to the role-assignment problem and internally non-selective with respect to $(\ell, \mathcal{S}_\ell)$. Suppose:*

- (E1) Outcome-profile neutrality. $\mathcal{S}_\ell$ is outcome-profile neutral for ℓ .*
- (E2) Declared environmental domain. The environmental system $\mathcal{E}$ and channel family Ch are part of the fixed target domain $\mathcal{D}_{\text{env}}$.*
- (E3) Admissible environmental anchor. The induced predicate $\text{Tr}_{\text{Ch}} : \mathcal{S}_\ell \rightarrow \{0, 1\}$ is stable, outcome-independent, and well-typed as an admissible environmental anchor.*
- (E4) Unique coupling asymmetry. There exists exactly one $s^* \in \mathcal{S}_\ell$ such that $\text{Tr}_{\text{Ch}}(s^*) = 1$.*

Then $X_{\text{env}}(\mathcal{S}_\ell) = s^$ and the assignment $R_{\text{env}} = R[\ell \mapsto s^*]$ is a unique admissible role-anchor assignment for ℓ on $\mathcal{D}_{\text{env}}$. Moreover, the selection is not an outcome-comparison-based, outcome-conditioned, or confirmation-based selector.*

Proof. By (E4) and Definition 5.2, $X_{\text{env}}(\mathcal{S}_\ell) = s^*$. So existence and uniqueness as a value of the environmental rule are immediate.

The forcing layer is supplied by Corollary 3.6: after internal non-selectivity and outcome-profile neutrality remove the internal route, and after indexical/provenance anchoring is not in play for the Class III case, the only remaining same-domain asymmetry locus is environmental coupling. It remains to show admissibility. By (E2), the environment and channel family are in the declared fixed

domain, so their use is not an undeclared same-scope import. By **(E3)**, Tr_{Ch} is stable under channel-preserving admissible transformations and outcome-independent. By Lemma 4.10, the declared coupling is an admissible construction anchor. By Lemma 4.11, every channel-preserving admissible transformation fixes s^* . Thus assigning ℓ to s^* commutes with the relevant identity-preserving transformations of $\mathcal{D}_{\text{env}}$ at the reference-fixing locus.

The assignment does not appeal to outcome comparison because **(E1)** makes all candidates outcome-profile neutral and because the rule consults only the declared boundary trace Tr_{Ch} . It does not appeal to confirmational success because outcome independence forbids dependence on favored constructed alternatives, posteriors, losses, utilities, or confirmational order. It does not appeal to an arbitrary selector because Definition 5.2 returns a value only in the singleton-coupling case. Therefore the assignment is a stable, bivalent, outcome-independent, same-domain role-anchor assignment. By the AASC import discipline in Definition 1.4, this is admissible role-anchor fixation rather than boundary selection or post hoc repair. Consequently, no admissible construction can avoid this outcome without violating fixed-domain invariance or introducing an inadmissible asymmetry source. $\square$

Corollary 6.2 (Fixation dichotomy). *For the exhausted Class III regime, role-anchor fixation is bivalent at the constructional-role level: either $\text{BTFix}_{\text{Ch}}(\ell, \mathcal{S}_\ell) = s^*$ for the unique singleton trace-positive candidate s^* , or no same-domain environmental fixation exists. There is no admissible intermediate status, partial preference, or later repair.*

Proof. This is immediate from the partial definition of BTFix_{Ch} , the singleton condition in Theorem 6.1, and the fail-closed cases in Lemma 5.4 and theorem 7.2. $\square$

Corollary 6.3 (Residual automorphism collapse). *Under the assumptions of Theorem 6.1, every admissible environmental automorphism fixes the role-realizer of ℓ . Therefore the residual automorphism group acts trivially on the environmental role-anchor assignment for ℓ .*

Proof. This is Lemma 4.11 applied to the unique selected token s^* . $\square$

Corollary 6.4 (Monotonicity under declared anchoring). *Passing from the internal domain $\mathcal{D}_{\text{int}}$ to the declared environmental domain $\mathcal{D}_{\text{env}}$ can only shrink the admissible automorphism group for the role-assignment problem:*

$$\text{Aut}_{\text{env}}(\mathcal{D}_{\text{env}}) \subseteq \text{Aut}_{\text{int}}(\mathcal{F}).$$

If the shrinkage leaves a singleton positive environmental orbit, reference is fixed for ℓ .

Proof. An environmental automorphism must preserve all internal admissibility, report, and outcome-profile structure, and must also preserve the channel package. Thus it satisfies all internal automorphism conditions plus additional constraints. The inclusion follows. If those constraints leave a singleton positive orbit, the selected token is fixed by Lemma 4.11 and the role anchor is fixed by Theorem 6.1. $\square$

Remark 6.5 (Relation to ordinary experimental practice). Ordinary experiments often record not merely an outcome value but also which detector, channel, memory bank, calibration line, or sensor interface produced the value. When that channel identity is part of the available report or interaction structure, the role assignment is anchored. The present theorem isolates the abstract condition under which that ordinary anchoring move is admissible even when the internal formal structure alone is non-selective.

7 Converse and no-go closure

The positive theorem is sharp only if the failed cases are closed. This section records the closure explicitly.

Definition 7.1 (Environmental failure modes). For a candidate set $\mathcal{S}_\ell$ and an alleged environmental role-anchor constraint, the failure modes are:

- C1. No declared channel.** $\mathcal{E}$ or Ch is not part of the fixed domain.
- C2. No positive coupling.** $\text{Tr}_{\text{Ch}}^{-1}(1) \cap \mathcal{S}_\ell = \emptyset$.
- C3. Non-unique positive coupling.** $|\text{Tr}_{\text{Ch}}^{-1}(1) \cap \mathcal{S}_\ell| \geq 2$.
- C4. Unstable coupling.** Coupling status is not preserved by identity-preserving channel-preserving transformation.
- C5. Outcome-conditioned coupling.** Coupling status depends on confirmational outcome, posterior, loss, utility, or downstream success.
- C6. Scalarized coupling.** A metric, probability, strength, ranking, or threshold not fixed in the domain is used to decide the selected token.
- C7. Delayed coupling repair.** The channel criterion is introduced, changed, or reinterpreted after reference failure while retaining the same evaluated act.

Theorem 7.2 (Full no-go: no same-scope environmental fixation without singleton stable coupling). *Let $\mathcal{F}$ be internally non-selective with respect to $(\ell, \mathcal{S}_\ell)$ and let the candidates be outcome-profile neutral. If no stable, outcome-independent, declared boundary trace has a unique positive value on $\mathcal{S}_\ell$, then no same-scope environmental constraint uniquely fixes the role anchor for ℓ .*

Proof. Assume toward contradiction that an environmental constraint uniquely fixes ℓ without satisfying the singleton stable coupling condition. Then one of the failure modes in Definition 7.1 must occur.

- C1.** If no declared channel is present, the alleged environmental selector is not typed in the fixed domain. Using it as authority is scope transport or undeclared domain reset, not same-scope fixation.
- C2.** If no candidate is positively coupled, the environment supplies no positive token. Selecting a token requires another selector.
- C3.** If several candidates are positively coupled, the environment leaves them tied. Selecting one requires a tie-breaker not supplied by Tr_{Ch} .
- C4.** If coupling is unstable, the role anchor can vary under an identity-preserving transformation. That is silent redescription, not role-anchor fixation.
- C5.** If coupling is outcome-conditioned, reference is fixed only after or by confirmational use. That is evidence-based symmetry breaking or post-selection repair, not prior admissible role-anchor fixation.
- C6.** If a scalar or ranking decides the choice without being fixed as part of the domain, the rule injects a boundary parameter. If it is fixed as part of the domain, the actual admissibility-relevant object is the bivalent predicate it induces; then the theorem reduces to the singleton case if unique, and fails otherwise.

C7. If the channel is introduced or reinterpreted after the failure while retaining the same evaluated act, this is delayed repair. AASC fixed-domain discipline permits explicit reset and fresh evaluation, but not same-act repair.

Every case either fails uniqueness, violates stability or outcome independence, imports a selector, or changes scope. Therefore no same-scope environmental constraint fixes reference without singleton stable coupling. $\square$

Corollary 7.3 (Symmetry-saturated converse). *If $\mathcal{S}_\ell$ is environmentally symmetry-saturated relative to Tr_{Ch} , then no admissible environmental rule can uniquely fix ℓ on the same fixed domain.*

Proof. Environmental symmetry saturation means that the boundary trace leaves no singleton positive candidate. Hence either **C2** or **C3** of Definition 7.1 occurs. The result follows from Theorem 7.2. $\square$

Corollary 7.4 (No post-selection environmental repair). *If no admissible boundary trace fixes ℓ at the construction time of the evaluated role-assignment act, then later success, Bayesian updating, posterior improvement, model refinement, downstream utility, calibration, future data, or reinterpretation cannot retroactively create standing for that same act. Any such update constitutes a new construction context and cannot repair the original role-anchor assignment. Such later material can only define a fresh evaluated domain or a fresh construction act.*

Proof. A role-anchor assignment must be fixed before the token is used as evidence. If the original domain lacks a singleton stable boundary trace, Theorem 7.2 gives fail-closed undefinedness for the environmental route. Later success or calibration can add information, but adding it changes the channel package, outcome ordering, or admissibility-relevant domain. Treating that later material as if it had fixed the earlier act is post-selection repair. AASC fixed-domain discipline permits explicit reset followed by fresh evaluation; it does not permit retroactive creation of standing for the same act. $\square$

Theorem 7.5 (Symmetry-saturated no-go for all admissible maps). *Suppose $\mathcal{S}_\ell$ is symmetry-saturated under channel-preserving identity transformations and no candidate has a unique positive coupling value. Then no admissible map*

$$Y : \mathcal{S}_\ell \rightarrow \mathcal{S}_\ell$$

fixes a unique occupant of ℓ on the same domain.

Proof. Let Y be admissible. Admissibility requires Y to commute with identity-preserving transformations on the fixed domain. Symmetry saturation gives, for any proposed selected token s , a channel-preserving transformation sending s to some $t \neq s$ while preserving all admissibility-relevant structure, or else leaves a non-singleton orbit of equally coupled candidates. If Y selects s , equivariance forces the transformed situation to select t under the same domain. Since the transformed situation is identity-equivalent to the original, uniqueness of s is not invariant. If Y refuses equivariance, it introduces a tie-break not preserved by the domain; that is silent redescription or parameter injection. Hence no admissible same-domain map selects uniquely. $\square$

Theorem 7.6 (Class III closure by exhaustion). *Within the fixed domain $\mathcal{D}_{\text{env}}$, every alleged Class III environmental role-anchor strategy falls into exactly one of the following outcomes:*

(K1) *singleton stable coupling, in which case Theorem 6.1 applies;*

- (K2) fail-closed undefinedness because the boundary trace is empty or non-unique on $\mathcal{S}_\ell$;
- (K3) illicit same-scope selector, parameter injection, or silent redescription;
- (K4) outcome-conditioned or post-selection repair; or
- (K5) explicit scope change followed by fresh evaluation on a new domain.

No sixth same-scope Class III route exists.

Proof. The alleged strategy either uses a declared stable bivalent boundary trace or it does not. If it does and the positive set is a singleton, (K1) holds. If it does and the positive set is not a singleton, (K2) holds by Definition 5.2. If it does not use a declared stable bivalent predicate, then the defect appears at one of the loci in Definition 7.1: no declared channel, instability, outcome-dependence, scalarization, or delayed repair. These are precisely (K3)–(K5) depending on whether the move claims same-scope authority, uses outcome-conditioned repair, or declares a new domain. Since Lemma 3.3 exhausts the admissible source forms, Table 1 maps alleged fourth sources back into those forms or into inadmissibility, and Definition 7.1 lists all ways the environmental boundary-trace conditions can fail, no further same-scope route remains. $\square$

Remark 7.7 (Why the converse does not forbid new experiments). The no-go results do not say that one may never add an environment, recalibrate an instrument, or collect new channel data. They say that doing so is a new domain or new evaluated act. The later successful domain may be admissible. What is forbidden is treating the later environmental enrichment as if it had already fixed reference for the earlier same-scope act.

8 Classification within the positive role-anchor construction space

The environmental route is Class III because the asymmetry that fixes the role-anchor is neither internal constructional structure, nor indexical/provenance anchoring, nor arbitrary selection. It is a declared boundary trace induced by a channel relation between candidate constructed tokens and an environment external to the internal formalism but declared in the target domain.

Definition 8.1 (Positive role-anchor constraint). A positive role-anchor constraint for $(\ell, \mathcal{S}_\ell)$ is a partial rule X such that, whenever $X(\mathcal{S}_\ell) = s^*$:

- (i) $s^* \in \mathcal{S}_\ell$;
- (ii) the selection is unique;
- (iii) the selection is stable under the admissible transformations of the declared domain;
- (iv) the selection is outcome-independent; and
- (v) the rule is fail-closed when its own admissibility conditions are not met.

Definition 8.2 (Class III environmental anchoring). A positive role-anchor constraint is *Class III environmental anchoring* iff its decisive invariant is a stable declared boundary trace $\text{Tr}_{\text{Ch}} : \mathcal{S}_\ell \rightarrow \{0, 1\}$ induced by environmental coupling and its output is given by X_{env} of Definition 5.2.

Proposition 8.3 (Class III is distinct from provenance and indexical anchoring). *Class III environmental anchoring is not reducible, as a formal matter, to provenance anchoring or indexical anchoring.*

Proof. A provenance anchor fixes reference by source, generation, or production history. Two candidate tokens may have the same internal provenance profile while only one remains coupled to a declared external channel. Then provenance does not distinguish them but environmental coupling does.

An indexical anchor fixes reference by centered position, self-location, or role-relative access. Two candidate tokens may be equally uncentered or equally inaccessible indexically while only one is environmentally coupled. Then indexical anchoring does not distinguish them but environmental coupling does.

Conversely, tokens may differ in provenance or indexical relation while having identical environmental coupling status. Therefore the environmental invariant is formally independent of those anchoring types. $\square$

Proposition 8.4 (Boundary-trace anchoring restores fixation for the anchored role). *If the assumptions of Theorem 6.1 hold, then admissible role-anchor fixation is restored for constructional role ℓ on the environmental domain $\mathcal{D}_{\text{env}}$.*

Proof. Failure of admissible role-anchor fixation for ℓ would require a non-trivial admissible reassignment of ℓ to a distinct candidate while preserving report content, outcome profiles, and the declared environmental boundary trace. But by Corollary 6.3, every admissible environmental automorphism fixes the unique coupled token s^* . So no such non-trivial reassignment remains for ℓ . $\square$

Remark 8.5 (Local restoration only). The restoration in Proposition 8.4 is local to the anchored role and the declared environmental domain. It does not imply that every other role in $\mathcal{L}$ is fixed, and it does not imply that the theory as a whole has eliminated every other form of reference underdetermination.

Closure-normal form

In current AASC terminology, the result has the following normal form:

- Kernel : Adm requires bivalent, fixed-domain, identity-preserving construction;
- Interface : ℓ requires a stable constructed-token occupant before confirmational use;
- Failure : Int and Idx are exhausted under non-selectivity;
- Forcing : Tr_{Ch} remains only after Int and Idx exhaustion;
- Output : BTFix_{Ch} either fixes the unique coupled token or fails closed.

This is why the paper no longer depends on the older “possible class member” framing. Class III is not merely permitted. In the specified exhausted regime, boundary-trace fixation is the only admissible same-domain route left.

9 Minimal executable construction witness

The witness keeps only the structure needed to instantiate the theorem.

Example 9.1 (Two-token environmental witness). Let

$$\mathcal{S}_\ell = \{s_A, s_B\}.$$

Assume:

- (i) s_A and s_B are outcome-profile indistinguishable;
- (ii) $\mathcal{F}$ is internally non-selective with respect to $(\ell, \mathcal{S}_\ell)$;
- (iii) an external environment $\mathcal{E}$ and channel family Ch are declared in $\mathcal{D}_{\text{env}}$;
- (iv) the induced boundary trace satisfies

$$\text{Tr}_{\text{Ch}}(s_A) = 1, \quad \text{Tr}_{\text{Ch}}(s_B) = 0;$$

- (v) Tr_{Ch} is stable and outcome-independent.

Then $X_{\text{env}}(\mathcal{S}_\ell) = s_A$ and $R_{\text{env}} = R[\ell \mapsto s_A]$ is the unique environmental role-anchor assignment.

Feature	s_A	s_B	Admissibility status	Selector consequence
Internal type	Same constructed-token type	Same constructed-token type	Admissible but non-selective	No internal type selector
Report content	Same role-compatible report profile	Same role-compatible report profile	Admissible but non-selective	No report-content selector
Outcome profile	$\text{Out}(R[\ell \mapsto s_A])$	Equal outcome profile under every admissible ordering	Admissible but non-selective	No outcome-profile selector
Indexical/-provenance marker	None declared as unique	None declared as unique	Absent as unique anchor	No indexical/provenance selector
Boundary trace	$\text{Tr}_{\text{Ch}}(s_A) = 1$	$\text{Tr}_{\text{Ch}}(s_B) = 0$	Admissible, declared, bivalent, singleton	Selects s_A
Rule status	Candidate survives trace	Candidate rejected by trace	Fail-closed partial rule	$X_{\text{env}}(\mathcal{S}_\ell) = s_A$

Table 2: Compact structural witness. The construction tokens are identical with respect to internal type, report content, outcome profile, and indexical/provenance status. The only admissibility-relevant asymmetry is the declared boundary trace, and its admissibility status is explicit.

Proposition 9.2 (Minimal construction witness: environmental coupling breaks non-selectivity). *The witness in Example 9.1 satisfies the assumptions of Theorem 6.1. Hence environmental coupling fixes a unique admissible role-anchor assignment for ℓ .*

Proof. Outcome-profile indistinguishability and internal non-selectivity supply the target problem. The declared environment and channel family place the boundary trace in the fixed domain. Stability and outcome independence supply admissibility. The equation $\text{Tr}_{\text{Ch}}(s_A) = 1$ and $\text{Tr}_{\text{Ch}}(s_B) = 0$ supplies unique coupling asymmetry. Therefore the environmental selection rule returns s_A and the main theorem applies. $\square$

Minimal construction executability

```
def X_env(candidates, Trace_Ch):
    coupled = [s for s in candidates if Trace_Ch(s) == 1]
    if len(coupled) == 1:
        return coupled[0]
    raise Exception("No unique boundary trace.")

# Minimal witness
```

```

S_E = ["s_A", "s_B"]

def Trace_Ch(s):
    return 1 if s == "s_A" else 0

assert X_env(S_E, Trace_Ch) == "s_A"

```

Remark 9.3 (What the executable witness proves). The code does not prove the physical truth of a channel relation. It is a minimal construction witness for the formal fail-closed rule: if the declared predicate has exactly one positive candidate, the rule returns that candidate; otherwise it fails closed. The proof of admissibility is supplied by the stability, outcome-independence, and fixed-domain constructed alternatives, not by the code itself.

10 Referee-facing objections and replies

Proposition 10.1 (Environmental coupling is not convention). *If Tr_{Ch} is stable under channel-preserving admissible transformations, then environmental selection is not merely a naming convention.*

Proof. A convention may choose one representative among symmetric alternatives without adding preserved structure. A stable boundary trace is different: it changes the admissible automorphism group by requiring channel preservation. If exactly one token is coupled, every channel-preserving admissible transformation fixes that token. The selected token is therefore fixed by invariant channel structure, not by a freely replaceable name. $\square$

Proposition 10.2 (Environmental coupling is not confirmation by another name). *Under Definition 4.7, environmental coupling cannot encode a confirmational preference among constructed alternatives.*

Proof. A confirmational preference depends on how evidence bears on constructed alternatives. Environmental coupling depends only on the declared channel relation. If changing posteriors, losses, constructed alternative rankings, or outcome success does not change Tr_{Ch} , then Tr_{Ch} cannot encode a confirmational preference. $\square$

Proposition 10.3 (Multiple coupled tokens do not yield partial reference). *If two or more candidates in S_ℓ satisfy $\text{Tr}_{\text{Ch}} = 1$, then Class III environmental anchoring does not yield partial reference, graded reference, or shared standing for ℓ on the same fixed role.*

Proof. The role ℓ is the fixed constructional role whose occupant is to be assigned. A non-singleton positive coupling set does not identify a unique occupant. Treating the several coupled tokens as partially or jointly occupying the same unique role introduces either a different role scheme, a quotient role, or a graded status. A different role scheme or quotient role is a scope change; a graded same-scope status is excluded by bivalence. Therefore the original unique-role question remains undefined rather than partially solved. $\square$

Proposition 10.4 (Delayed calibration is fresh evaluation). *If calibration performed after a reference failure creates a unique environmental channel, the resulting success is a new evaluated act on a new or enriched domain, not a repair of the prior same-scope failure.*

Proof. Calibration changes the channel package used to type Tr_{Ch} . If the prior act lacked that package or lacked uniqueness under it, the prior act had no Class III fixation. Adding calibration later may create a new admissible domain, but treating it as if the original act already had that reference would change the identity conditions of the earlier act. That is same-act repair or silent redescription, both forbidden under the fixed-domain discipline. $\square$

Proposition 10.5 (Externality does not imply unrestricted authority). *The fact that $\mathcal{E}$ is external to the internal constructional formalism of $\mathcal{C}$ does not make every environmental fact admissibility-relevant.*

Proof. Only declared, stable, outcome-independent channel facts enter $\mathcal{D}_{\text{env}}$. Environmental facts that are not part of the channel package, are not preserved by the admissible transformations, or are consulted only because they select a desired result are not admissible environmental anchors. They are either irrelevant background, scope changes, or illicit selector imports. $\square$

Proposition 10.6 (The environment itself is not the selector). *A Class III selector is admissible only when it is trace-mediated. Replacing the declared trace by open-ended appeal to the environment destroys same-domain admissibility.*

Proof. By Lemma 4.4, the admissibility-relevant object is Tr_{Ch} , not unrestricted environmental reality. Open-ended environmental appeal can always introduce further distinctions among candidates; unless those distinctions are fixed in $\mathcal{D}_{\text{env}}$, they are extra-domain preferences. If they are fixed, they become part of the trace/domain package and the problem is a new or enriched domain. Hence the environment itself is not the selector in the original same-scope theorem. $\square$

Table 3: Attack surface and formal response.

Attack	Failure diagnosis	Hardened response
“Just choose the coupled-looking one.”	Hidden selector if the channel predicate is not declared and stable.	Require $\mathcal{E}, \text{Ch}$ in $\mathcal{D}_{\text{env}}$ and require stability.
Two tokens are coupled.	Non-unique positive set.	$X_{\text{env}} = \uparrow$; no tie-break.
No token is coupled.	No positive candidate.	$X_{\text{env}} = \uparrow$; no argmax-over-zero selection.
Use coupling strength.	Parameter injection unless the threshold is declared.	Reduce to declared bivalent predicate; otherwise fail closed.
Use the token that confirms best.	Outcome-conditioned selection.	Excluded by outcome independence.
Add calibration after failure.	Post-selection repair or scope change.	Fresh evaluation only; no same-act repair.
Rename the coupled token.	Silent redescription if coupling status changes.	Channel-preserving automorphisms must preserve Tr_{Ch} .

Attack	Failure diagnosis	Hardened response
Treat environment as an oracle.	Externality without fixed channel.	Only declared channel invariants count.

11 Scope limits and non-claims

- N1.** The theorem does not derive the existence of an external environment. It states what follows if such an environment and channel are part of the declared domain.
- N2.** The theorem does not prove that environmental coupling confirms either H_0 or H_1 . It fixes the role whose report may later be used in confirmation.
- N3.** The theorem does not license arbitrary causal, semantic, pragmatic, or methodological selection. Only stable, outcome-independent, bivalent channel coupling is admissible.
- N4.** The theorem does not repair a prior failure of reference. It permits explicit domain reset and fresh evaluation.
- N5.** The theorem does not collapse all forms of role-anchor fixation into environmental coupling. It identifies Class III as one positive route among distinct anchoring classes.
- N6.** The theorem does not make a scalar theory of coupling strength. Scalar quantities are admissible only as already-declared bookkeeping that induces the bivalent channel predicate used by the rule.
- N7.** The theorem is local to the role ℓ and candidate set $\mathcal{S}_\ell$. Other roles may remain underdetermined.

Remark 11.1 (Counterexample burden). A counterexample must be same-scope. It must preserve internal non-selectivity, outcome-profile neutrality, the declared environmental channel package, outcome independence, bivalence, and identity-preserving transformation while still producing a unique environmental selection different from X_{env} . A proposal that changes the channel, adds a selector, consults outcomes, or changes the role scheme is not a counterexample to the theorem; it is a different domain or a failed same-domain move.

12 Conclusion

Boundary-trace fixation is a forced fixed-domain construction result in the exhausted Class III regime. When internal constructional structure and outcome profiles do not select a unique constructional-role occupant, and when indexical/provenance anchoring is unavailable, a declared external channel is the only remaining same-domain route if it contributes a stable, outcome-independent, singleton invariant on the candidate set. The resulting role-anchor assignment is admissible because it is fixed before confirmational use, preserved under the transformations of the environmental domain, and fail-closed outside the singleton case.

The theorem also explains why the condition is sharp. Multiple coupled candidates, absent coupling, unstable coupling, posterior-conditioned coupling, scalar tie-breaking, and delayed calibration do not produce same-scope Class III fixation. They either fail uniqueness, become selector smuggling,

import new boundary parameters, change scope, or attempt post-selection repair. Class III therefore occupies a precise location in the positive fixation space: stable environmental channel invariants can fix role-anchors, but only as declared, bivalent, non-confirmational boundary traces.

A AASC source dependency ledger

This appendix records the AASC machinery used by the proof architecture. The present paper is self-contained at the local theorem level; the ledger identifies the source spine that motivates each admissibility constraint.

Table 4: AASC import ledger.

AASC source	Imported structural role	Use in this paper
<i>Admissible Reference Fixation as a Necessary Condition for Non-Trivial Theory-Level Empirical Confirmation</i>	Admissible assignments; admissible automorphisms; no symmetry-respecting selector; no evidence-based interface selection; anchoring restores role-anchor fixation. Labels: AASC:RF:def:admissible, AASC:RF:def:aut, AASC:RF:prop:nosel, AASC:RF:prop:noevidencesel, AASC:RF:prop:anchoring.	Provides the role-assignment architecture and explains why environmental channel information can be an anchoring constraint rather than a confirmational tie-break.
<i>The Structure of Admissibility</i>	Reference-stable invariants, AMetric boundary, no selector, bivalence, no graded legitimacy, gate uniqueness, standing conservation. Labels include AASC:AG:def:reference-stable-invariant, AASC:AG:thm:ametric-boundary, AASC:AG:prop:no-selector, AASC:AG:thm:bivalence, AASC:AG:cor:no-graded, AASC:AG:prop:gate-uniqueness, AASC:AG:thm:conservation.	Supplies fail-closed partiality, bivalent coupling status, no scalar tie-breaks, and stability under identity-preserving transformations.
<i>Minimal Conditions for Admissible Construction</i>	Fixed-domain admissibility, standing, identity-preserving construction, and no retroactive standing creation.	Used to discipline the distinction between declared environmental domain and post hoc environmental repair.
<i>Standing-Admissibility Identity Closure</i>	No second same-domain classifier, route exhaustion, and governance-role collapse. Labels include AASC:SC:thm:unique-same-domain-gate-classification, AASC:SC:cor:no-second-same-domain-admissibility-bearing-classifier.	Blocks the move from coupling status to an additional same-scope classifier beyond the bivalent boundary trace.

AASC source	Imported structural role	Use in this paper
<i>A Fixed-Domain Exhaustion Theorem</i>	Fixed-domain/same-scope comparison class, fail-closed typing, no boundary selector smuggling, structural exhaustion. Labels include <code>AA SC:FDE:def:fixed-domain-class</code> , <code>AASC:FDE:def:fail-closed-typing-rule</code> , <code>AASC:FDE:lem:definable-selector-smuggling</code> , <code>AASC:FDE:thm:exhaustion</code> .	Provides the closure form used in Theorem 7.6.
<i>Closure by Exhaustion for Same-Scope Operators under Admissibility</i>	Same-scope closure logic and no surviving totalization/family repair route.	Used as the template for the Class III closure-by-exhaustion theorem.
<i>No Post-Selection Repair and No Deferred Admissibility</i>	Post hoc selection, repair, and delayed admissibility cannot recover same-act standing.	Blocks outcome-conditioned environmental selection and delayed calibration as same-scope repairs.
<i>Prohibition of Silent Re-description</i>	Reclassifying the same act under altered identity conditions is inadmissible.	Blocks unstable coupling and after-the-fact channel reinterpretation.
<i>Non-Factorizability of Admissibility</i>	Admissibility is global with respect to standing-bearing commitments and cannot be assembled from independently admissible parts without checking interactions.	Blocks treating independent environment and token admissibility as sufficient unless the channel relation itself is admitted on the fixed domain.
<i>Admissible Record Construction in Physically Non-Selective Quantum Regimes</i>	Non-selective record construction, selector taxonomy, boundary/quotient/representation locus exhaustion. Labels include <code>def:selector-locus</code> , <code>thm:selector-locus-trichotomy</code> , <code>thm:selector-exhaustion</code> .	Provides the selector-exhaustion posture for constructed-token reference under physical or internal non-selection.

B Failure-class alignment

The environmental failure modes of Definition 7.1 align with the standard AASC failure classes as follows.

Table 5: Environmental failures and AASC failure classes.

Environmental failure	AASC failure class	Reason
No declared channel	Scope transport / ill-typed import	The environmental fact is not part of the fixed domain.
No positive coupling	Undefinedness / no selector	The boundary trace supplies no selected candidate.

Environmental failure	AASC failure class	Reason
Non-unique positive coupling	Boundary selector smuggling	Any further choice among coupled candidates requires extra same-scope selection.
Unstable coupling	Silent redescription	Coupling status changes under identity-preserving transformation.
Outcome-conditioned coupling	Deferred validation / post-selection repair	The role-anchor assignment depends on downstream confirmational status.
Scalarized coupling	Parameter injection	A metric, score, or ranking performs boundary work unless reduced to a declared bivalent predicate.
Delayed coupling repair	Same-act repair or explicit reset	Later channel enrichment cannot retroactively fix the earlier act.

References

- [1] Amos Jay Maley. *Admissible Reference Fixation as a Necessary Condition for Non-Trivial Theory-Level Empirical Confirmation*. AASC source corpus, 2026.
- [2] Amos Jay Maley. *The Structure of Admissibility: Ametricity, Bivalence, and the Unique Admissible Interior*. AASC source corpus, 2026.
- [3] Amos Jay Maley. *Minimal Conditions for Admissible Construction: Admissibility, Standing, and Identity-Preserving Construction*. AASC source corpus, 2026.
- [4] Amos Jay Maley. *Standing–Admissibility Identity Closure: A Fixed-Domain Meta-Necessity Closure*. AASC source corpus, 2026.
- [5] Amos Jay Maley. *A Fixed-Domain Exhaustion Theorem*. AASC source corpus, 2026.
- [6] Amos Jay Maley. *Closure by Exhaustion for Same-Scope Operators under Admissibility*. AASC source corpus, 2026.
- [7] Amos Jay Maley. *Admissible Record Construction in Physically Non-Selective Quantum Regimes*. AASC source corpus, 2026.